

Certificate Request Form

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To access Zapways OTA API, OTA's are required to create a key-pair PKI certificate and provide Zapways with the public certificate to load into the airlines system. Once loaded the OTA will then present the certificate and key in the message payload.

The certificate can be a self-signed certificate and needs to adhere to X.509 version 3 standard. Should be valid for at least one (1) year and up to five (5) years, and should not be re-created once it is created while valid.

CREATING A CERTIFICATE:

1. You will need to download OpenSSL
2. Create a private key using the following command

```
openssl genrsa -out key.pem 2048
```

3. Create a Self-Signed certificate Using the following command

```
openssl req -new -x509 -key key.pem -out cert.pem -days 365 -outform PEM
```

4. Use the following Details

(C) Country Name	**AF**
(S) State or Province Name	**KABUL**
(L) Locality Name	**AFN**
(O) Organization Name	**NEW LINK TOURIST & TRAVEL AGENCY**
(OU) Organizational Unit Name	OTA
(CN) Common Name	**1951**
Email Address	***ismail@newlinkaf.com***

Send the created certificate in CERT.PEM format to sales.asia@zapways.com. Once installed, we will send a secure link to the email address provided certificate. This will contain the OTA documentation, Keys and Test Script.

Below are the instructions create a Certificate that you will need to implement in your code and pass to Zapways every time a connection is established. Instructions have been provided in JAVA and .NET.

JAVA

Java applications expect to retrieve certificates from a Java Key Store (JKS). JVM (JDK) comes with a utility called 'Keytool' to help you create a new key store. However, 'Keytool' does not let you import an existing private key for which you already have a certificate. You need to do this yourself and here's how.

1. First you will need to convert the certificates. Using the following command, convert the private key (key.pem to key.der)

```
openssl pkcs8 -topk8 -nocrypt -in key.pem -inform PEM -out key.der -outform DER
```

2. Then you will need to convert the PEM cert into DER

```
openssl x509 -outform der -in cert.pem -out cert.der
```

3. Combine the Private and Public Keys. g the following command (*Use the ImportKey.class to combine both key.der and cert.der)*

```
java ImportKey key.der cert.der
```


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4. This will create a jks file (without jks extension) in the following location “C:\Documents and Settings\User Name\keystore.ImportKey” using 'importkey' as alias and 'importkey' as password.

DOT NET

.NET applications expect to retrieve certificates from a PFX. OpenSSL can be used, here's how:
Then you can export the certificate as follows

1. Using the following command, export the certificate:

```
openssl pkcs12 -export -out cert.pfx -inkey key.pem -in cert.pem
```